Project Apache Tika

Task: Detect document type

Scenario:
Tika as a content analysis framework, it can extract text or metadata. But before extraction, it must first identify what kind of file it is dealing with. This is necessary because different formats require different parsers. Tika can detect multiple document types, such as pdf, word, xml, and more. For this it uses different techniques for document types detection. Your task is to explore the architectural design decisions that achieve document type detection in Apache Tika.

Existence

Question 1: What are the architectural components and connectors in Apache Tika that achieves document type detection? Briefly name and describe each component, determine their type (physical or logical), determine how they are connected, and how they achieve the requirements.

Property

Question 2: What solutions (e.g., patterns or tactics) are implemented to support quality attributes in Apache Tika for achieving document type detection? Mention the considered quality attributes and describe the solutions applied, and how they achieve the requirements.

Executive

Question 3: Which tools and technologies are integrated to handle Apache Tika document type detection? Mention the technologies and tools, their associated components, rationale, and how they achieve the requirements.

Project Apache Tika

Task: Extensibility requirement

Scenario:
Tika as a content analysis framework needs to parse text from documents with different formats. However, new document types appear constantly. Therefore, Apache Tika needs to support the extensibility of its type detector and parser. Your task is to explore the architectural design decisions that achieve the extensibility requirements in Apache Tika.

Existence

Question 1: What are the architectural components and connectors in Apache Tika that support (affect) its extensibility requirement for its detector and parser? Briefly name and describe each component, determine their type (physical or logical), determine how they are connected, and how they achieve the requirements.

Property

Question 2: What solutions (e.g., patterns or tactics) are implemented to support the extensibility quality attribute in Apache Tika for its detector and parser? Mention the considered quality attributes and describe the solutions applied, and how they achieve the requirements.

Executive

Question 3: Which tools and technologies are integrated to achieve its extensibility quality attribute in Apache Tika for its detector and parser? Mention the technologies and tools, their associated components, rationale, and how they achieve the requirements.